DAY 12 TASK

Missing Value Identification

Complete Submission Report | Dataset • Python Code • Output • Findings

1. Objective

Identify missing values in the dataset, calculate their counts and percentages, and document the results without deleting or imputing values.

2. Dataset Details

Dataset file: titanic.csv | Rows: 891 | Columns: 12

3. Python Code

```
import pandas as pd

# Load titanic.csv (keep it in the same folder as this script)
df = pd.read_csv("titanic.csv")

print("Dataset shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head().to_string(index=False))

missing_count = df.isnull().sum()
missing_percentage = (missing_count / len(df)) * 100

missing_summary = pd.DataFrame({
    "Missing Values": missing_count,
    "Missing Percentage": missing_percentage.round(2)
})
missing_summary = missing_summary[missing_summary["Missing Values"] > 0]
missing_summary = missing_summary.sort_values(
    by="Missing Values", ascending=False
)

print("\nMissing Value Summary:")
print(missing_summary.to_string())

missing_summary.to_csv(
    "missing_value_summary.csv",
    index=True,
    index_label="Column"
)
print("\nSaved: missing_value_summary.csv")
```

4. Execution Output

```
Dataset shape: (891, 12)\n\nFirst 5 rows:
 PassengerId  Survived  Pclass                                                Name    Sex  Age  SibSp  Parch
           1         0       3                             Braund, Mr. Owen Harris   male 22.0      1      0
           2         1       1 Cumings, Mrs. John Bradley (Florence Briggs Thayer) female 38.0      1      0
           3         1       3                              Heikkinen, Miss. Laina female 26.0      0      0 STON/O2
           4         1       1        Futrelle, Mrs. Jacques Heath (Lily May Peel) female 35.0      1      0
           5         0       3                            Allen, Mr. William Henry   male 35.0      0      0

\nMissing Value Summary:\n          Missing Values  Missing Percentage
Cabin                687              77.10
Age                  177              19.87
Embarked               2               0.22
```

Column	Missing Values	Missing Percentage
Cabin	687	77.10%
Age	177	19.87%
Embarked	2	0.22%

5. Findings

• Cabin: 687 missing values (77.10%).

• Age: 177 missing values (19.87%).

• Embarked: 2 missing values (0.22%).

• Missing values may affect analysis and model performance. They should be handled based on the dataset and analysis goal, not automatically deleted.

6. Conclusion

The analysis identifies columns containing missing data and quantifies the extent of missingness. No values were deleted or filled; this report documents the missing-value profile so a suitable treatment can be chosen later.

7. Submission Checklist

✓ Objective and dataset details included
✓ Python code included
✓ Actual dataset preview and calculated output included
✓ Missing-value summary table included
✓ Findings and conclusion included

Note: Counts in this report are calculated directly from the included titanic.csv file. The summary CSV and code are also provided separately if the submission portal requests them.